Synthetic Control

ECON726

Tara Sullivan

Tara Sullivan
tara.sullivan55@login.cuny.edu

October 28, 2025

Housekeeping

Homework:

- Regression discontinuity designs next week (MHE Chp 6)
- Replication assignment due! Remember: we want primary results!

Empirical project

Paper selection: Due **September 22** [5% of final grade]

- Done!

Project plan: Due **October 14** [15% of final grade]

- Done!

Replication results: **November 3**: [10% of final grade]

- You should submit some of your main replication results. The final results should be formatted as tables and charts in a word document or PDF, submitted on Brightspace. Your code and data should accessible on GitHub.

Presentation: Due **December 2** & **December 9** [10% of final grade]

- Prepare a 5-8 minute presentation on your paper. Focus on the research question, main result, and one extension.

Final Paper: Due **December 16** [35% of final grade]

- Your paper should contain 6-10 pages of text, and 3-6 pages of figures (double spaced, 12 point font). You must provide the code and data to replicate your findings. Your final project should be submitted on Brightspace, and your code and data should be uploaded to github (note: if your data is too large to upload to github, let me know).

Outline

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Diff-in-diff review (Card and Kruegr, 1994)

Note: these notes are from my PhD course on Causal Inference, taught by Kaspar Wüthrich

Set-up:

- Two groups (treatment and control): $G = 1$ and $G = 0$
- Two periods (before and after the treatment): $T = 0$ and $T = 1$
- Only one group is treated in one period

Data:

- Survey of 400 fast food stores both in NJ and PA both before and after the minimum wage increase in NJ

Results:

- Positive effect of increasing the minimum wage on employment

Formal analysis of simple DiD model

Potential outcomes framework:

- D: treatment indicator; $D = T \cdot G$
- $Y(d)$: potential outcome
- Y: observed outcome; $Y = D \cdot Y(1) + (1 - D) \cdot Y(0)$:

Key assumption:
Common trends: $\mathbf{E}[Y(0)|G, T = 1] - \mathbf{E}[Y(0)|G, T = 0]$ *does not depend on* G

Under common trends assumption, average treatment effect on the treated defined as:

$$\begin{aligned}\Delta_{DiD} =& \mathbf{E}[Y(1) - Y(0)|G = 1, T = 1] \\ =& (\mathbf{E}[Y|G = 1, T = 1] - \mathbf{E}[Y|G = 1, T = 0]) \\ & - (\mathbf{E}[Y|G = 0, T = 1] - \mathbf{E}[Y|G = 0, T = 0])\end{aligned}$$

Estimation can proceed based on the simple regression model:

$$Y = \alpha + \gamma G + \lambda T + \Delta_{DiD} D + \epsilon$$

Discussion and notes about diff-in-diff model

- Common trends assumption requires that average outcomes for treated and controls would have followed *parallel trends* in the absence of treatment
- Assumption of common trends is fundamentally untestable. We can get some idea of its plausibility when several periods are available, by checking the pre-trends are the same for treated and controls
- One can also run placebo tests, and include leads into a regression-based DiD model, to check the plausibility of the parallel trends assumption

Let's move on to issues and challenges with diff-in-diff

Issue #1: common trends and scaling

The parallel trends assumption is **scale dependent**

- Parallel trends won't hold both for Y and a monotone function of Y (like $\log(y)$) at the same time

How to overcome this issue:

- Answer: Changes-in-changes model; see Athey and Imbens (2006)

https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0262.2006.00668.x

> *We... propose an estimator that can be applied using either repeated cross section or panel data. Our approach provides an estimate of the entire counterfactual distribution of outcomes that would have been experienced by the treatment group in the absence of the treatment and likewise for the untreated group in the presence of the treatment.*

Issue #2: fuzzy designs

Simple diff-in-diff assumes $D = T \cdot G$. Often, $D \neq T \cdot G$. Example: Duflo (2001)

- Looking at primary school construction in Indonesia, to measure the returns to education
- Many primary schools were constructed in districts with few schools ($G = 1$), and few new schools were constructed in districts with a lot of schools ($G = 1$)
- Primary school completion rate increases more in $G = 1$ than in $G = 0$, but not from 0% to 100%

How to overcome this issue:

- Answer: fuzzy differences-in-differences; see de Chaisemartin and D'Haultfoeuille (2018)

https://academic.oup.com/restud/article-abstract/85/2/999/4096388

Issue #3: choice of control group

Simple diff-in-diff model assumes a control group with common trends exists

How do we choose a control group if there are multiple options?

- Synthetic control!

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Synthetic Control

Athey and Imbens (2017)

> *Synthetic control is "arguably the most important innovation in the policy evaluation literature in the last 15 years." (pg. 3)*

First developed in Abadie and Gardeazabal (2003)

- Study effect of terrorism on aggregate income

Formalized in Abadie, Diamond, and Hainmueller (2010)

- Impact of Prop 99 on smoking in California

Formalized in the comparative politics literature in Abadie, Diamond, and Hainmueller (2014)

- Impact of German reunification on West Germany

Example motivating synthetic control: Mariel boatlift

Card (1990): Do inflows of immigrants depress wages and the employment of natives in local labor markets?

Context: Mass immigration of Cubans from Cuba's Mariel Harbor to the US between April and October 1980

- In 1980, Fidel Castro announced anyone wishing to leave Cuba could do so
- 125,000 Cubans emigrated to Florida over six months
- Increased the Miami labor force by 7%

Viewed as a natural experiment; Mariel boatlift provided an exogenous shift in labor supply

- Used individual-level data CPS data on unemployment from four comparison cities (Atlanta, LA, Houston, Tampa)
- Claims cities are comparable based on demographics and economic conditions
- Estimated diff-in-diff model

Outcome: results were contrary to competitive labor-market model

> *The Mariel influx appears to have had virtually no effect on the wages or unemployment rates of less-skilled workers, even among Cubans who had immigrated earlier*

Critique of Card (1990)

Empirical critiques of Card (1990):

- Choice of control was subjective
- Standard errors reflect sampling variance, as opposed to uncertainty about the ability of the control group to reproduce the counterfactual of interest

Solution: Synthetic control

Abadie and Gardeazabal (2003)

Study effect of terrorism in Basque Country on aggregate income

- In the early 1970's, Basque Country was one of the richest regions in Spain
- ETA was founded to promote the establishment of independent Basque state in Spain. Became a violent separatist group
- Thirty years of terrorist and political instability had adverse economic consequences; but how would the economy have involved in the absence of violence?
- Relative to the rest of Spain in the 1960's, Basque Country had higher per capita income, higher investment ratio, was more densely populated, had a higher percentage of industrial production, and a better educated labor force

  $\Longrightarrow$ Parallel trends would not hold
- Generate a synthetic Basque that most closely resembles the real Basque country before terrorism (in the pre-treatment period)

https://www.aeaweb.org/articles?id=10.1257/000282803321455188

Set-up

- $J + 1$ units (countries or states), indexed by i
  - $i = 1$ is the treated state
  - J states in the "donor pool"
- $T = T_0 + T_1$ periods
  - $T_0 > 0$ pre-treatment periods
  - $T_1 > 0$ post-treatment
- $i = 1$ is treated after T_0; all other J units serve as potential controls
- $\{Y_{it}(0), Y_{it}(1)\}$ are potential outcomes of unit i at time t
- $D_{it} = 1$ if $i = 1$ and $t > T_0$; $D_{it} = 0$ otherwise

Synthetic control potential outcomes

Causal effect of interest:

$$\alpha_i = Y_{it}(1) - Y_{it}(0), \quad i = 1, t = T_0 + 1, \ldots, T$$

Can relate observed outcome to potential outcomes:

$$Y_{it} = Y_{it}(0) + D_{it}\alpha_{it},$$

Since we're only interested in $i = 1$:

$$\begin{aligned} Y_{1t} &= Y_{1t}(0) + D_{1t}\alpha_{1t} \\ \alpha_{1t} &= \underbrace{Y_{1t}(1)}_{\text{observed}} - \underbrace{Y_{1t}(0)}_{\text{unobserved}}, \quad t = T_0, \ldots, T \end{aligned}$$

Because we're only interested in unit $i = 1$, and because $Y_{1t}(1)$ is observed, we only need to estimate $Y_{1t}(0)$ (the unobserved counterfactual)

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Synthetic Control

Abadie, Diamond, and Hainmueller: Synthetic Control Methods for Comparative Case Studies

Concern about health risks lead to the passage of Proposition 99 in California in 1988

- First modern tobacco control program in the United States
- Created cigarette excise tax of 25 cents per pack, and earmarked the tax revenue for health and anti-smoking causes
- Following its passage, laws were enacted prohibiting smoking in restaurants, workplaces
- Largely viewed as having successfully cut smoking in California

Question: What was the effect of Prop 99 on cigarette consumption in California?

- **Problem:** California received the treatment. What control can you compare California to?

Finding a control

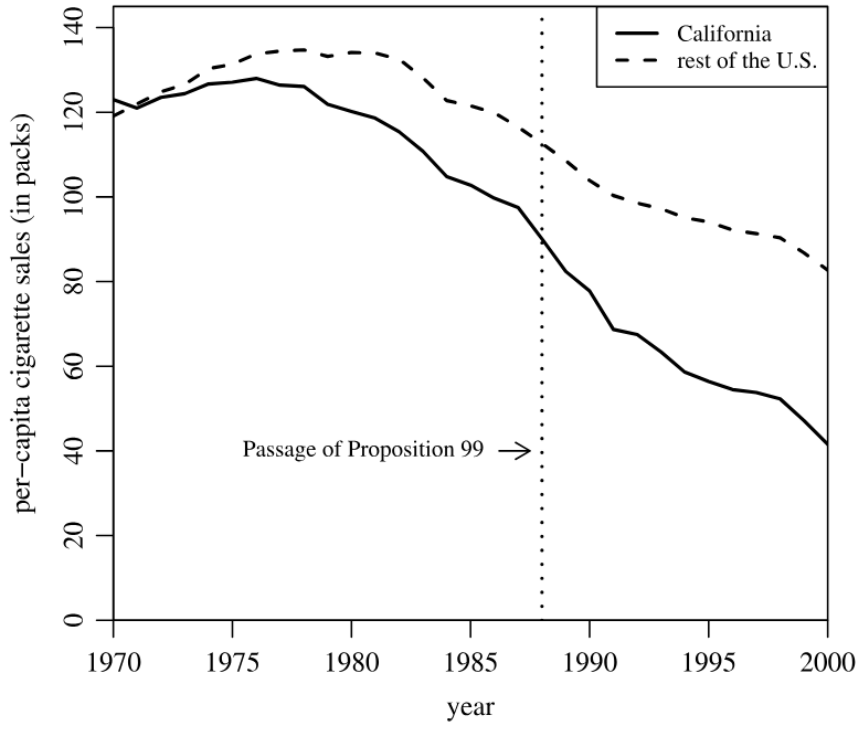


Figure 1. Trends in per-capita cigarette sales: California vs. the rest of the United States.

The rest of the United States may not provide a suitable comparison group for California

- Before the ban, cigarette consumption in California differed from the rest of the US

Building a synthetic California

Create a donor pool of states from which to build a synthetic California. Include all states that:

- Did not adopt large-scale tobacco control programs during the sample period
- Did not raise cigarette taxes by 50 cents or more in the post period

Construct a synthetic California as a weighted average of potential control states

- Weights should be chosen so that the resulting synthetic California best reproduces the values of a set of predictor of cigarette consumption in California before the passage of Prop 99

State weights in synthetic California

Table 2. State weights in the synthetic California

State	Weight	State	Weight
Alabama	0	Montana	0.199
Alaska	–	Nebraska	0
Arizona	–	Nevada	0.234
Arkansas	0	New Hampshire	0
Colorado	0.164	New Jersey	–
Connecticut	0.069	New Mexico	0
Delaware	0	New York	–
District of Columbia	–	North Carolina	0
Florida	–	North Dakota	0
Georgia	0	Ohio	0
Hawaii	–	Oklahoma	0
Idaho	0	Oregon	–
Illinois	0	Pennsylvania	0
Indiana	0	Rhode Island	0
Iowa	0	South Carolina	0
Kansas	0	South Dakota	0
Kentucky	0	Tennessee	0
Louisiana	0	Texas	0
Maine	0	Utah	0.334
Maryland	–	Vermont	0
Massachusetts	–	Virginia	0
Michigan	–	Washington	–
Minnesota	0	West Virginia	0
Mississippi	0	Wisconsin	0
Missouri	0	Wyoming	0

Cigarette sales predictor means

Table 1. Cigarette sales predictor means

Variables	California		Average of 38 control states
	Real	Synthetic	
Ln(GDP per capita)	10.08	9.86	9.86
Percent aged 15–24	17.40	17.40	17.29
Retail price	89.42	89.41	87.27
Beer consumption per capita	24.28	24.20	23.75
Cigarette sales per capita 1988	90.10	91.62	114.20
Cigarette sales per capita 1980	120.20	120.43	136.58
Cigarette sales per capita 1975	127.10	126.99	132.81

NOTE: All variables except lagged cigarette sales are averaged for the 1980–1988 period (beer consumption is averaged 1984–1988). GDP per capita is measured in 1997 dollars, retail prices are measured in cents, beer consumption is measured in gallons, and cigarette sales are measured in packs.

Synthetic California

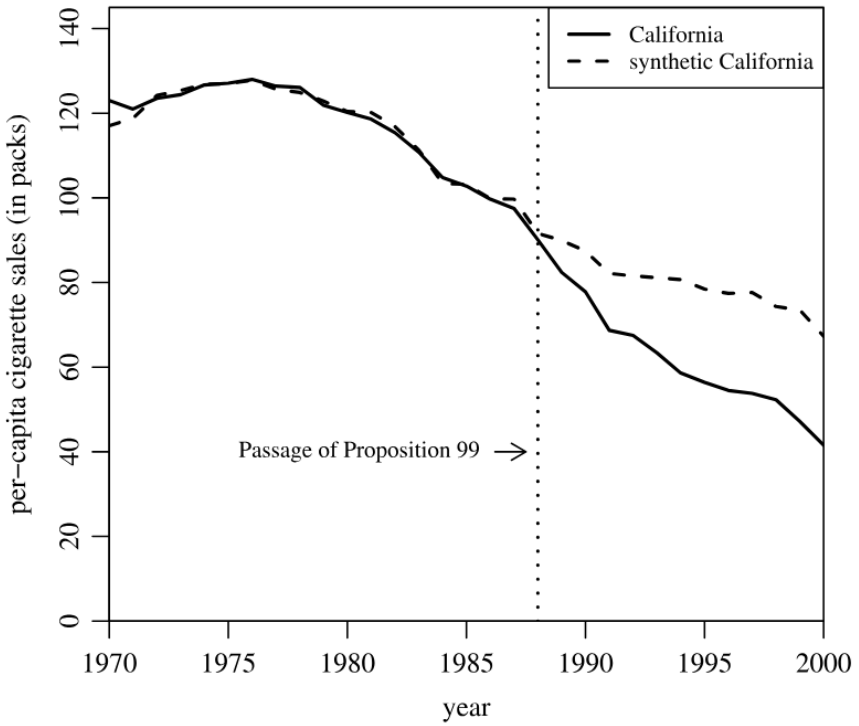


Figure 2. Trends in per-capita cigarette sales: California vs. synthetic California.

Code available: https://github.com/tara-sullivan/econ726-notebooks/blob/main/notebooks/09-synthetic_control.ipynb

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Defining synthetic $Y_{1t}(0)$

Synthetic control group defined as weighted average of units in the donor pool, and can be represented by a vector of weights $W = (w_2, \ldots, W_{J+1})$, where

- $0 \leq w_i \leq 1$ (No unit receives a negative weight, but units can receive a 0 weight)
- $\sum_{i=2}^{J+2} w_i = 1$

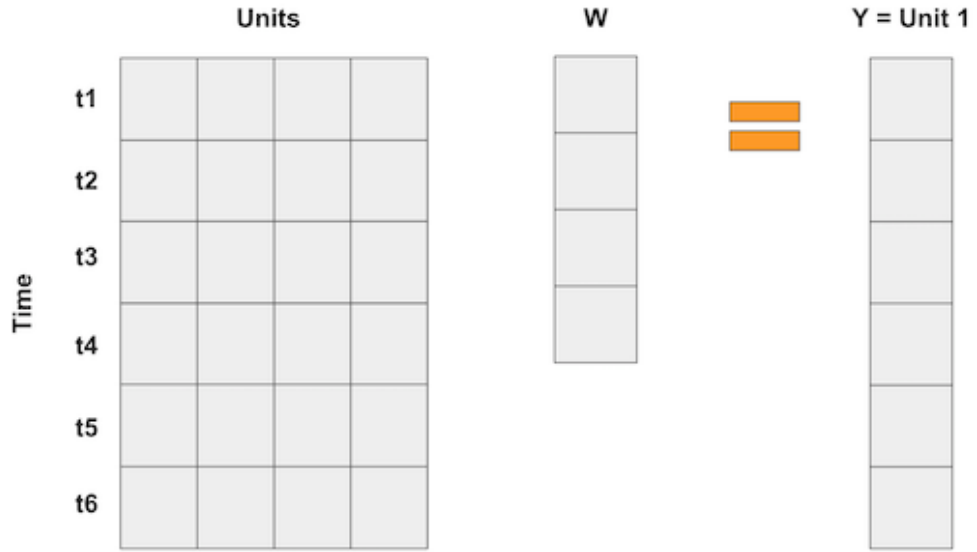
Then, for $t = T_0 + 1, \ldots, T$:

$$\hat{Y}_{1t}(0) = \sum_{j=2}^{J+1} w_j Y_{jt}$$

Understanding weights

Then, for $t = T_0 + 1, \ldots, T$:

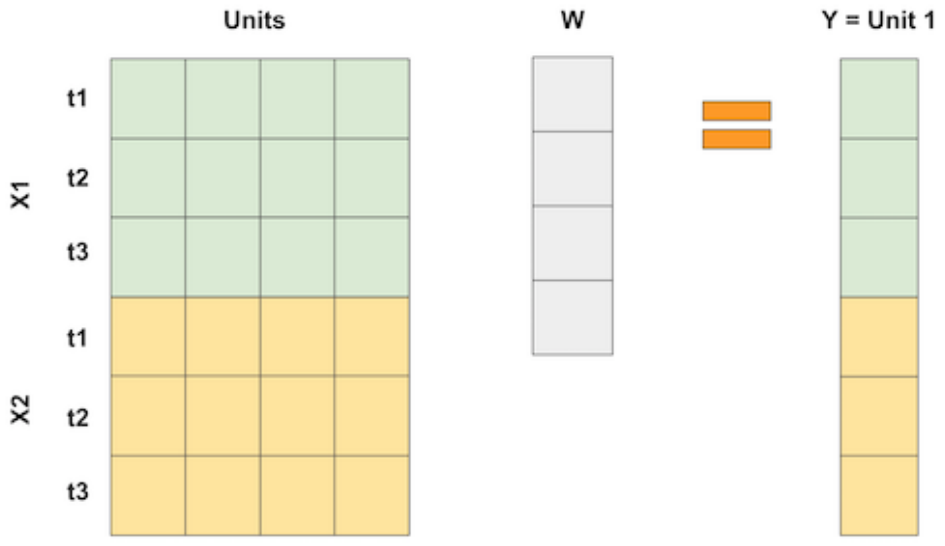
$$Y_{1t}(0) = \sum_{j=2}^{J+1} w_j Y_{jt}$$



Understanding weights

For $t = T_0 + 1, \ldots, T$:

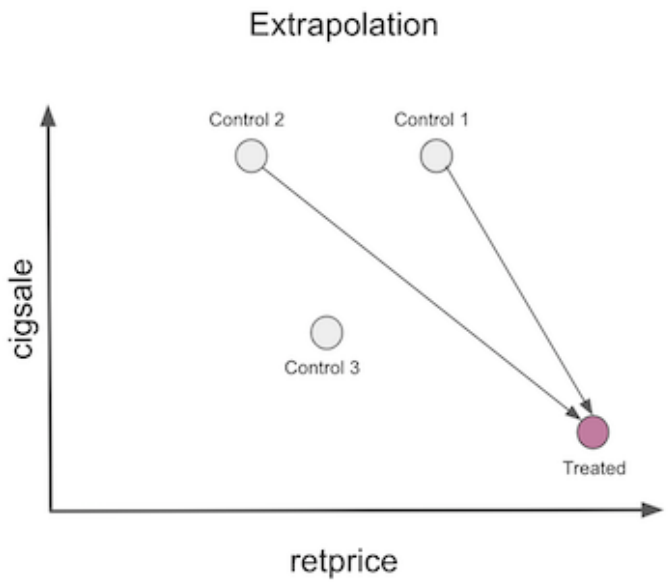
$$Y_{1t}(0) = \sum_{j=2}^{J+1} w_j Y_{jt}$$

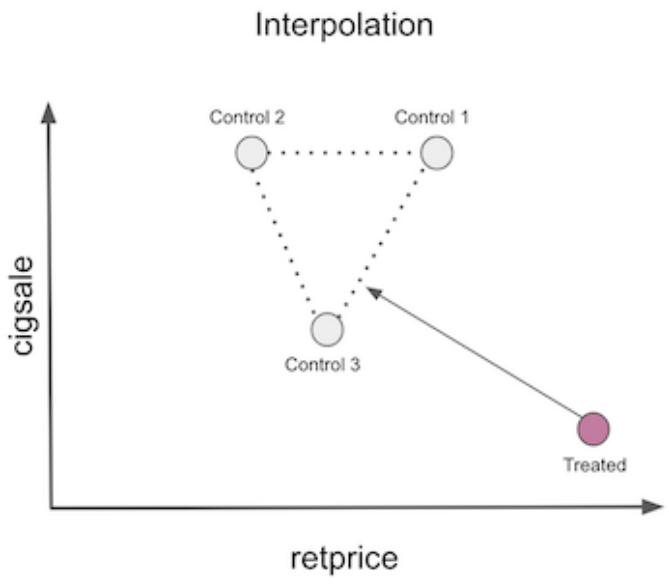


What if we tried using linear regression for this task?

Interpolation vs extrapolation

Why $0 \leq w_i \leq 1$ and $\sum_{i=2}^{J+2} w_i = 1$





Estimation

Let:

- X_1: $k \times 1$ vector of pre-intervention characteristics for $i = 1$
- X_0: $k \times J$ matrix with the same characteristics for the control unit

Abadie et al. (2010) propose to estimate the synthetic control group such that the distance between pre-treatment characteristics and synthetic control group are minimized:

$$W^* = \arg\min_W ||X_1 - X_0 W||$$

$$\text{subject to} \quad 0 \leq w_i \leq 1$$

$$\sum_{i=2}^{J+1} w_j$$

for some norm $||\cdot||$ (note: a norm is just distance, but in a multivariate space)

Weights W can be found by minimizing:

$$||X_1 - X_0 W|| = \sqrt{(X_1 - X_0 W)' V (X_1 - X_0 W)}$$

for some symmetric positive definite matrix $k \times k$ matrix V. More intuitively, because V is positive definite, its diagonal has values $v1, \ldots, v_k$, and this minimization problem looks like:

$$\sum_{m=1}^{k} v_m \left(X_{1m} - \sum_{i=2}^{J+1} w_i X_{im} \right)^2$$

where v_m reflects the relative importance we assign to variable m when we measure the discrepancy between the treated unit and the synthetic control.

This class is not about numerical optimization, consider examples

Differences-in-differences review

Synthetic Control

Example: Abadie, Diamond, and Hainmueller (2010)

Estimating synthetic controls

Inference for synthetic controls

Inference

Could the results from synthetic control be driven by chance?

- How often would we obtain results of this magnitude if we had chosen a state at random instead of California?

To answer this, run **placebo tests**

- Apply synthetic control to states that **did not** implement a large-scale tobacco control program during the sample period
- If placebo studies estimate effects similar to the one estimated for the treated unit, then the analysis does not provide significant evidence of a negative effect of Proposition 99 on cigarette sales in California

Placebo effects

Calculate placebo effect for region $i > 1$ at time t

$$\hat{\alpha}^*_{it} = \underbrace{Y_{it}(1)}_{\text{Pretend like } i \text{ is treated}} - \underbrace{Y_{it}(0)}_{\text{Calculate synthetic control for } i\text{, as if it were treated}}$$

For each placebo effect for each i (including the actually treated $i = 1$), you can calculate the root mean squared error in the pre-intervention period:

$$RMSE_{i,pre} = \sqrt{\frac{1}{T_0}\sum_{t=0}^{T_0}\left(Y_{it} - \sum_{j=1}^{J+1} w_j^* Y_{jt}\right)^2}$$

- For unit i, the weights to generate synthetic i are given by w_j^* (for $i = j$, $w_j = 0$)

Can also calculate RMSE for each region in the post-intervention period:

$$RMSE_{i,post} = \sqrt{\frac{1}{T - T_0}\sum_{t=T_0+1}^{T}\left(\hat{\alpha}^*_{it}\right)^2}$$

P-value

Then take the ratio of these errors:

$$p_i = \frac{RMSE_{i,post}}{RMSE_{i,pre}}$$

- $RMSE_{i,pre}$: How different synthetic i is from real i in the pre-period. We want this small
- $RMSE_{i,post}$: How different synthetic i is from real i in the post-treatment period. We expect this to be larger than $RMSE_{i,pre}$

Rank ratios from greatest to smallest, and see if the true treated groups p-value is large, relative to the placebo effects